A sharp cutoff for transient Hall defects in graph complements

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Research note for focused review, 21 October 2026

Question. Under bounded complement codegree, how large can a graph be if a Hall obstruction to zero-idleness triangle equality disappears under universal-vertex padding? The proposed answer is an exact order cutoff. This note isolates the structural argument from the longer algorithm paper. Priority remains unresolved.

Let G be a connected finite simple graph and $H = \overline{G}$, with $|N_H(u) \cap N_H(v)| \le k$ for $u \ne v$, where $k \ge 2$. For $xy \in E(G)$ orient $a = d_G(x) \le b = d_G(y)$ and set $d = b - a$. Write $A = N_G(x) \setminus N_G(y)$, $B = N_G(y) \setminus N_G(x)$, $C = N_G(x) \cap N_G(y)$ and $R = V \setminus (N_G(x) \cup N_G(y))$; their sizes are α, β, c, r. Then

$$a = \alpha + c, \quad b = \beta + c, \quad d = \beta - \alpha, \quad n = a + \beta + r, \quad r \le k.$$

For uniform measures on *open* neighborhoods, $\kappa_0(x, y) \le c/b$. Cancel common mass and multiply by ab: each source in A supplies b, each source in C supplies d, and each target in B demands a. Weighted Hall gives equality exactly when every target set has enough adjacent supply. A deficient set $S \subseteq B \setminus \{x\}$ has $s = |S|$ and excludes p sources of A and q of C, with positive numerator

$$F = (s + p - \beta)a + d(p + q) > 0.$$

Call its leading coefficient $L = s + p - \beta$ its *slope*: adjoining a universal vertex to G increases F by L.

Theorem 1 (Cutoff)**.** *Every deficient set with $L < 0$ satisfies $F \le k^2 + k + 2 - n$, hence $n \le k^2 + k + 1$. There is no such set for $k = 2$. For every $k \ge 3$ the order bound is attained. Above this order every defective edge satisfies $\beta \le \max(k + 1, 2k - 2)$.*

Proof. Every source sees target x, and source y sees all targets, so a deficient set excludes x and has $s \ge 2$. An excluded source and x have S as common H-neighbors, so $s \le k$. Two targets have x and all excluded sources as common H-neighbors, giving $p + q \le k - 1$. Also $\alpha \ge p + 1$ and $a \ge \alpha + q$. If $p \ge 2$, two excluded exclusive sources share $S \cup \{y\}$ in H, hence $s \le k - 1$.

Put $e = p + q$, $u = a - e \ge 1$ and $h = \beta - s - p \ge 1$. The deficit identity becomes $F = e(s + p - \alpha) - hu > 0$. Integrality and $\alpha \ge p + 1$ imply

$$hu \le e(s - 1) - 1, \qquad u + h \le hu + 1 \le e(s - 1), \qquad a + \beta \le (e + 1)s + p.$$

For $p \le 1$ the last expression is at most $k^2 + 1$; for $p \ge 2$ it is at most $k^2 - 1$. More precisely, putting $N = k^2 + k + 1$,

$$N + 1 - n - F = \big[N - ((e + 1)s + p + r)\big] + e(\alpha - p - 1) + (u - 1)(h - 1) \ge 0.$$

For $k = 2$, $hu \le e(s - 1) - 1 \le 0$ is impossible. Above the cutoff every deficient set has $L \ge 0$, so $\beta \le s + p \le \max(k + 1, 2k - 2)$. □

*Proposed attribution subject to the named author's acceptance. Substantial Codex/Astra assistance; no independent human mathematical review is claimed.

Proposition 1 (Boundary refinement, candidate)**.** *Let $k \geq 3$, $n = k^2 + k + 1$, and suppose an edge has a deficient set with $L < 0$. Then $r = s = k$, $p = 1$, $q = k - 2$, $\alpha = 2$, $F = 1$, $b = k^2 - 1$, and exactly one of*

$$(a, \beta, L) = (k, k^2 - k + 1, -k(k-2)), \qquad (k^2 - k - 1, k + 2, -1)$$

holds. Moreover $\kappa_0(x, y) = c/b - 1/(ab)$, and adjoining one universal vertex restores triangle equality at xy. These are numerical types, not uniqueness of the whole graph.

Proof outline. Equality in the cutoff chain forces $p = 1$, $e = k - 1$, $s = r = k$, $\alpha = p + 1 = 2$, $F = 1$, and $(u - 1)(h - 1) = 0$ with $uh = k(k - 2)$. This gives the two types and $b = k^2 - 1$. Let $A = \{y, z\}$. Since x, z already share all k vertices of S in H, z sees every vertex of R in G and every target outside S. For each $s \in S$, the vertices y, s already share z in H, so not all k vertices of R can be H-neighbors of s. Thus z reaches s in two G-steps through R. Common-source exceptions use a two-step path through y, so all residual costs are at most two. Every positive Hall witness is transient because $p \leq 1$ and $\beta \geq k + 2$, hence its numerator is one by the same equality analysis. Maximum cost-one flow therefore leaves exactly one unit for cost two. After padding, each such numerator becomes $1 + L \leq 0$. All other nonempty target sets have nonpositive slope: for $|S| \leq k$ this follows from $p \leq 1$ and $\beta \geq k + 2$; larger sets exclude no source. Their nonpositive numerators cannot increase; the empty set has deficit zero before and after padding. □

Realizations of both types. Take x, y, z and disjoint sets S, W, R, T, P, with $|S| = |R| = k$, $|W| = k - 2$, and put exactly

$$x \sim S \cup T, \qquad z \sim S \cup \{y\}, \qquad W \sim S, \qquad R \sim \{x, y\}$$

in H; vertices of P are isolated in H. Choose $(|T|, |P|) = (k(k-2), 0)$ or $(1, k(k-2) - 1)$. Both complements are connected and have pair codegree at most k. More generally, taking $P = \varnothing$ and $1 \leq |T| \leq k(k-2)$ attains $F = k^2 + k + 2 - n$ at every integer $3k + 2 \leq n \leq k^2 + k + 1$. The set S gives the asserted parameters. Adding an H-edge inside T in the first case, or inside P in the second, gives nonisomorphic examples; the numerical classification does not assert graph rigidity.

What is established background and what needs review. Weighted Hall, common-mass cancellation, and transport duality are classical. Li–Lu give unequal-degree weighted Hall criteria for a related LLY equality bound [2]. Chen–Liu–You supply the complement framework and local counting principles [1, Section 5]; their four-block template [1, Section 6] also underlies these constructions, with a modification of $k - 2$ complement edges. Asai–Yamada study zero-idleness triangle equality for specific edge-deleted complete graphs [3]. The proposed contribution to assess is the universal sharp cutoff and its boundary refinement, not those ingredients.

Verification scope. The longer paper supplies computational checks. Lean files connect actual graph sets to the cutoff, scalar consequences, boundary counts and paths. Weighted Hall and its graph-specific integer clone-transport interpretation are also checked. Metric common-mass cancellation, the real/integer transport bridge, extremal realization and the complete curvature corollary remain outside the formal proof. This note remains subject to adversarial and expert review.

[1] K. Chen, S. Liu, Z. You, *Criteria on forbidden subgraphs in the complements for positive Lin–Lu–Yau curvature*, 2026. https://arxiv.org/abs/2605.03470.

[2] Y. Li, L. Lu, *Ricci Curvature Formula: Applications to Bonnet–Myers Sharp Irregular Graphs*, 2024, Corollaries 2–3. https://arxiv.org/abs/2409.15667v2.

[3] Y. Asai, T. Yamada, *Ollivier Ricci curvature on graphs obtained by removing edges from complete graphs*, 2026. https://arxiv.org/abs/2605.30830v2.